GavinKlorfine–Project1

Gavin Klorfine

2026-02-17

Question 1

Minnesota High School Graduates: The `Hoyt` data in vcdExtra gives a $4 \times 3 \times 7 \times 2$ table classifying nearly 14000 graduates by (a) post-high school `Status`, (b) `Rank` in their graduating class, (c) father's `Occupational` status (7 levels, from 1=High to 7=Low) and (d) `Sex`. `Status` is considered the outcome variable. How does it depend on the other factors?

Part (a)

Analyze and display the associations between status and each of the other variables individually. In doing this, you might also consider focusing attention on the distinction between College and Non-college Status.

```r
data("Hoyt", package = "vcdExtra")

(status_occup <- (margin.table(Hoyt, c("Status", "Occupation")) |> assocstats()))
```

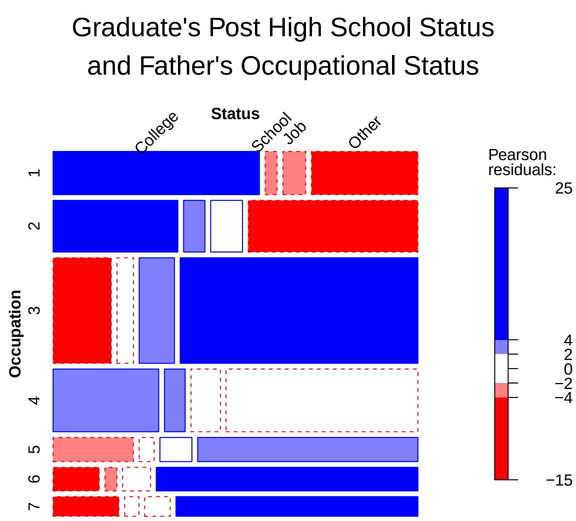
```
##                     X^2 df P(> X^2)
## Likelihood Ratio 1425.8 18        0
## Pearson          1479.1 18        0
##
## Phi-Coefficient   : NA
## Contingency Coeff.: 0.309
## Cramer's V        : 0.188
```

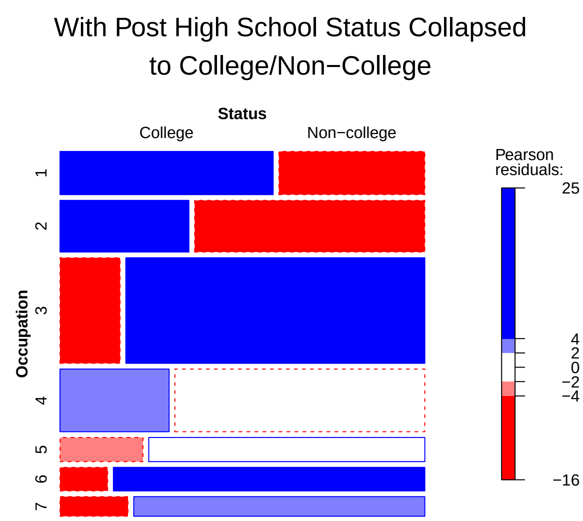
```r
# Analyze college vs non-college
distinct_status <- collapse.table(
  Hoyt, Status = c("College", "Non-college", "Non-college", "Non-college")
)

(
  dist_status_occup <- (
    margin.table(distinct_status, c("Status", "Occupation")) |> assocstats()
  )
)
```

```
##                     X^2 df P(> X^2)
## Likelihood Ratio 1336.5  6        0
```

```
## Pearson          1392.4  6        0
##
## Phi-Coefficient   : NA
## Contingency Coeff.: 0.301
## Cramer's V        : 0.316
```

```
mosaic(
  ~ Occupation + Status, data = Hoyt,
  shading = TRUE, gp = shading_Friendly,
  rot_labels = c(top = 45),
  main = "Graduate's Post High School Status\nand Father's Occupational Status"
)
mosaic(
  ~ Occupation + Status, data = distinct_status,
  shading = TRUE, gp = shading_Friendly,
  main = "With Post High School Status Collapsed\nto College/Non-College"
)
```





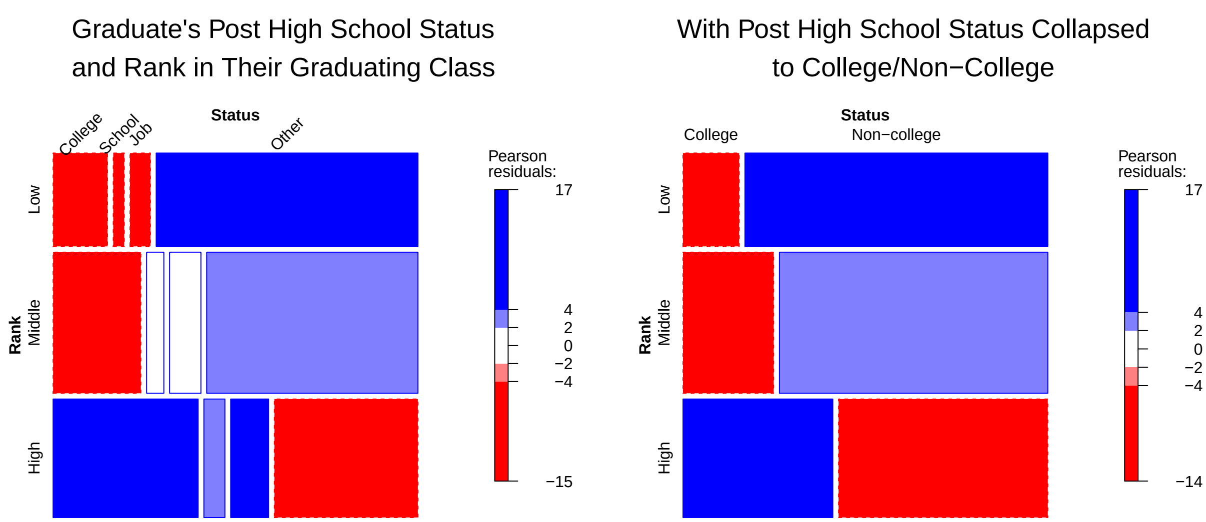

```
(status_rank <- (margin.table(Hoyt, c("Status", "Rank")) |> assocstats()))
```

```
##                     X^2 df P(> X^2)
## Likelihood Ratio 1062.6  6        0
## Pearson          1046.0  6        0
##
## Phi-Coefficient   : NA
## Contingency Coeff.: 0.264
## Cramer's V        : 0.194
```

```
(
  dist_status_rank <- (
    margin.table(distinct_status, c("Status", "Rank")) |> assocstats()
  )
)
```

```
##                        X^2 df P(> X^2)
## Likelihood Ratio 740.33  2        0
## Pearson          734.63  2        0
##
## Phi-Coefficient   : NA
## Contingency Coeff.: 0.224
## Cramer's V        : 0.229
```

```r
mosaic(
  ~ Rank + Status, data = Hoyt,
  shading = TRUE, gp = shading_Friendly,
  rot_labels = c(top = 45),
  main = "Graduate's Post High School Status\nand Rank in Their Graduating Class"
)
mosaic(
  ~ Rank + Status, data = distinct_status,
  shading = TRUE, gp = shading_Friendly,
  main = "With Post High School Status Collapsed\nto College/Non-College"
)
```




```r
(status_rank <- (margin.table(Hoyt, c("Status", "Sex")) |> assocstats()))
```

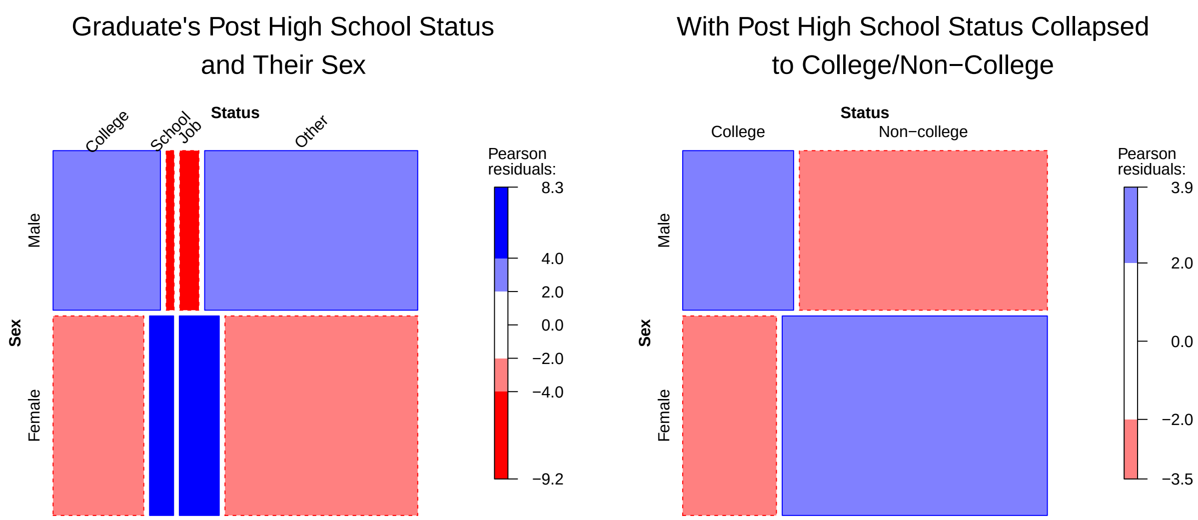
```
##                        X^2 df P(> X^2)
## Likelihood Ratio 361.55  3        0
## Pearson          341.87  3        0
##
## Phi-Coefficient   : NA
## Contingency Coeff.: 0.155
## Cramer's V        : 0.156
```

```r
(
  dist_status_rank <- (
    margin.table(distinct_status, c("Status", "Sex")) |> assocstats()
```

```
  )
)
```

```
##                         X^2 df   P(> X^2)
## Likelihood Ratio 38.816  1 4.6581e-10
## Pearson          38.928  1 4.3980e-10
##
## Phi-Coefficient   : 0.053
## Contingency Coeff.: 0.053
## Cramer's V        : 0.053
```

```
mosaic(
  ~ Sex + Status, data = Hoyt,
  shading = TRUE, gp = shading_Friendly,
  rot_labels = c(top = 45),
  main = "Graduate's Post High School Status\nand Their Sex"
)
mosaic(
  ~ Sex + Status, data = distinct_status,
  shading = TRUE, gp = shading_Friendly,
  main = "With Post High School Status Collapsed\nto College/Non-College"
)
```



Part (b)

Consider and analyze the associations among the predictor variables: `Rank`, `Occupation` ***and*** `Sex`. ***Is there anything of interest worth mentioning or analyzing in further detail? Is it useful to consider collapsing some of the categories?***

```
# Associations among predictors
assocstats(margin.table(Hoyt, c("Rank", "Occupation")))
```

```
##                     X^2 df P(> X^2)
## Likelihood Ratio 172.19 12        0
## Pearson          172.07 12        0
##
## Phi-Coefficient   : NA
## Contingency Coeff.: 0.11
## Cramer's V        : 0.078
```

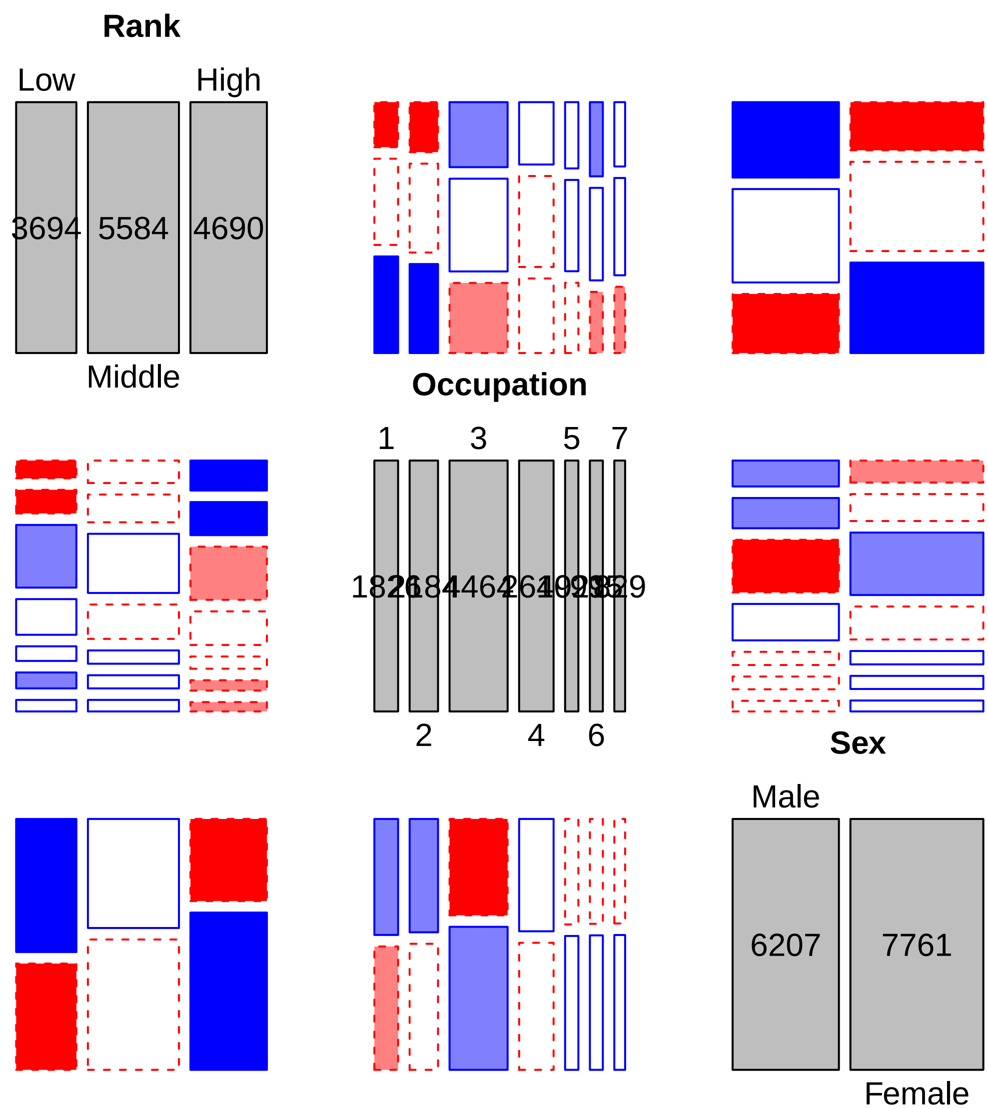
```
assocstats(margin.table(Hoyt, c("Rank", "Sex")))
```

```
##                     X^2 df P(> X^2)
## Likelihood Ratio 387.27  2        0
## Pearson          384.32  2        0
##
## Phi-Coefficient   : NA
## Contingency Coeff.: 0.164
## Cramer's V        : 0.166
```

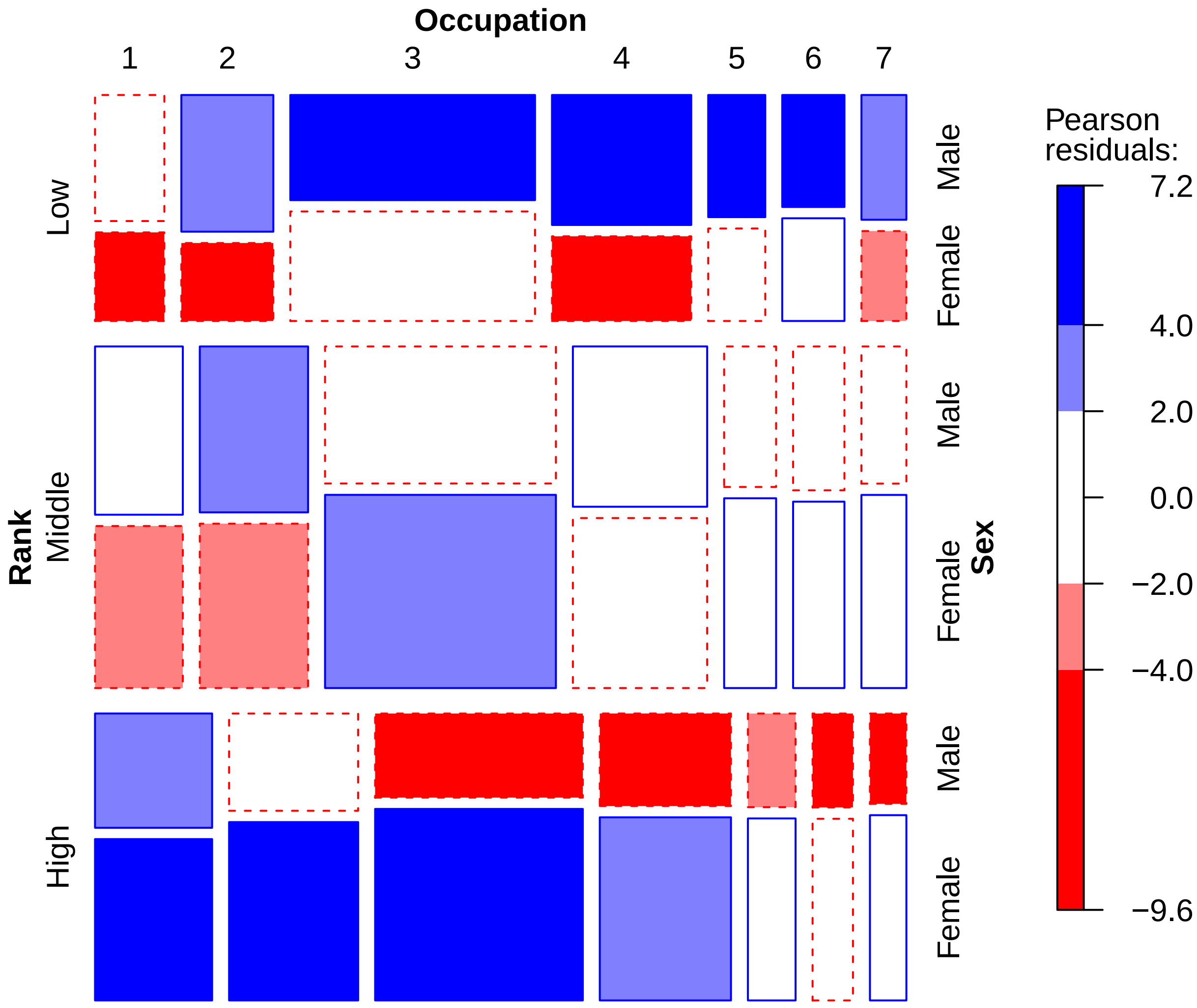
```
assocstats(margin.table(Hoyt, c("Sex", "Occupation")))
```

```
##                     X^2 df   P(> X^2)
## Likelihood Ratio 58.269  6 1.0105e-10
## Pearson          58.139  6 1.0738e-10
##
## Phi-Coefficient   : NA
## Contingency Coeff.: 0.064
## Cramer's V        : 0.065
```

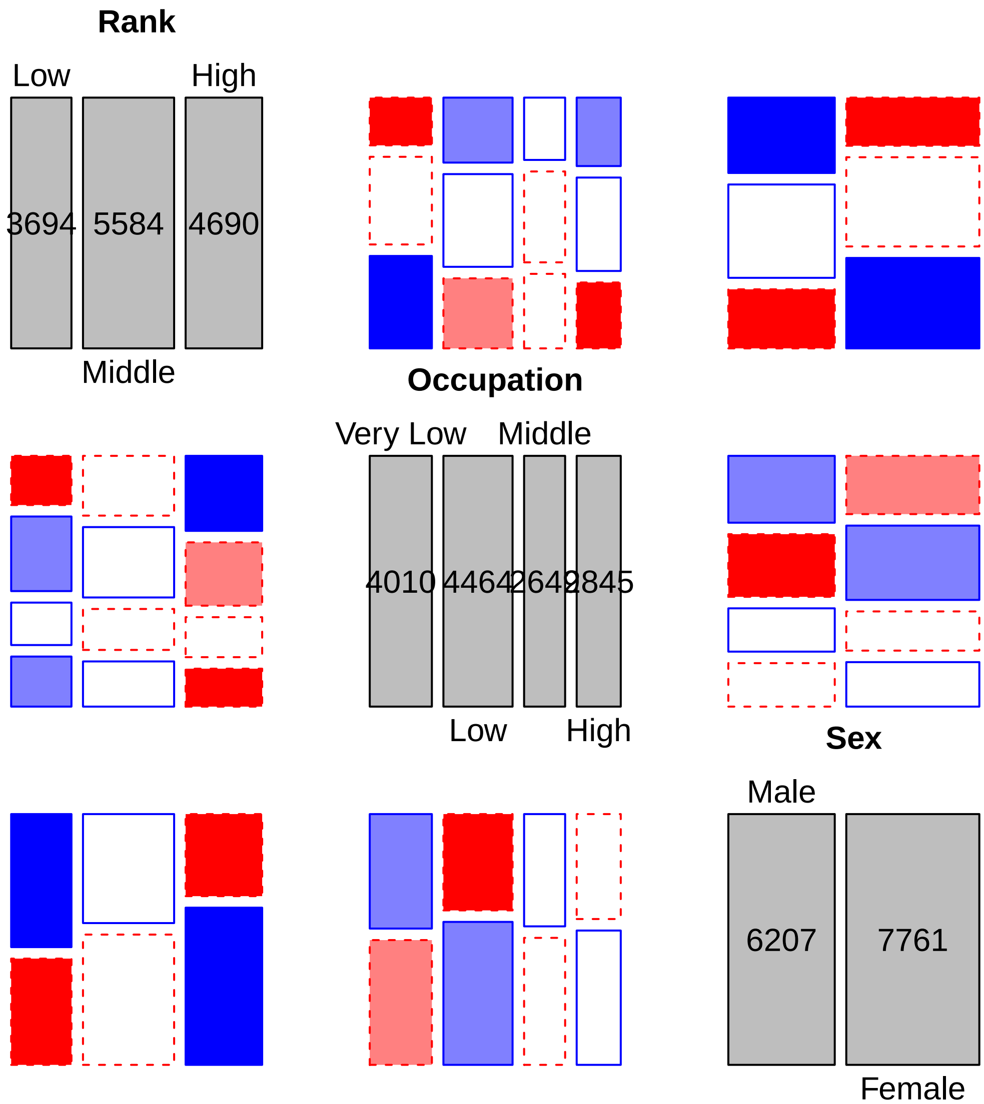
```
# Display associations
pred_tab <- margin.table(Hoyt, c("Rank", "Occupation", "Sex"))

pairs(pred_tab, gp = shading_Friendly)
```



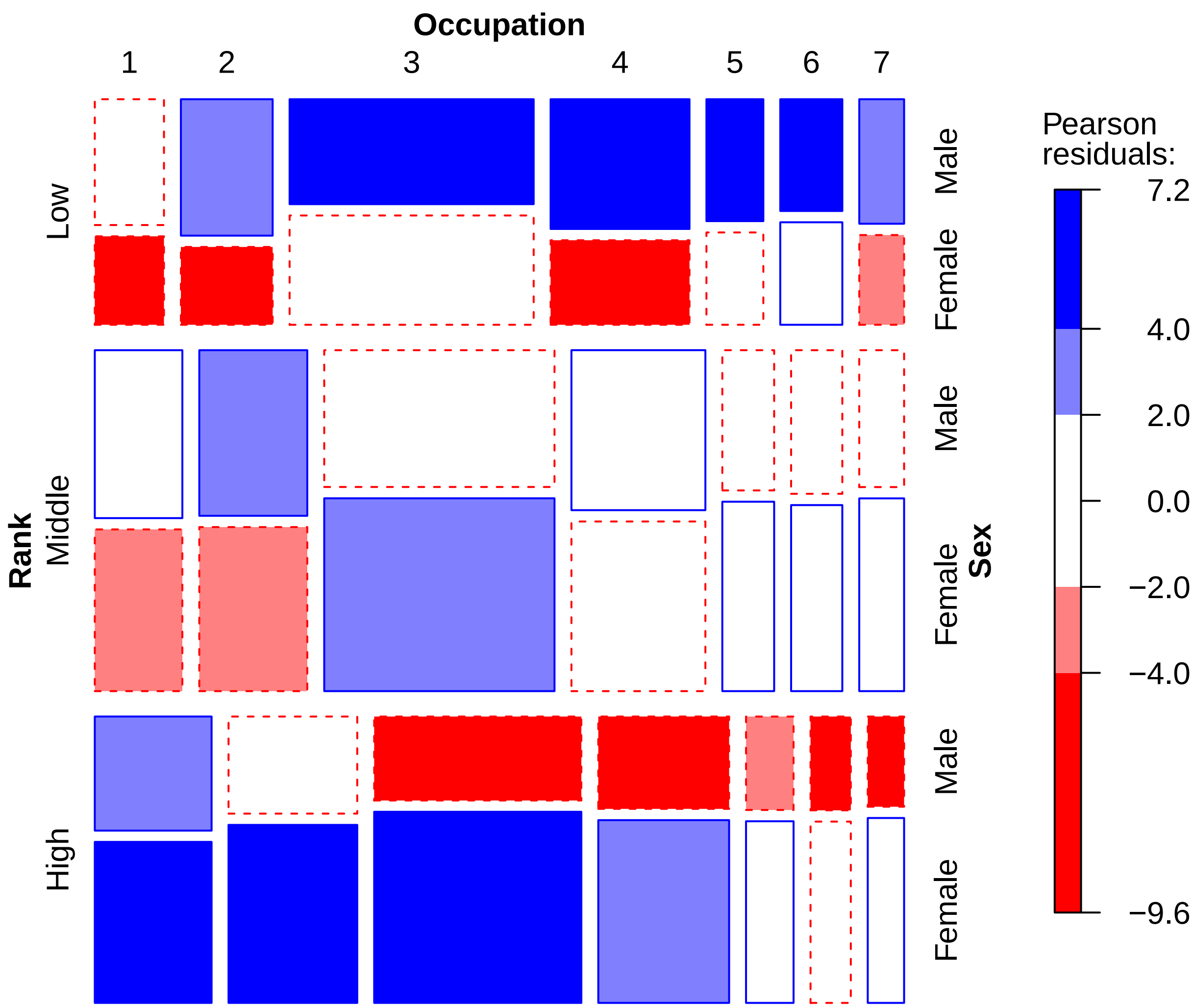

```r
mosaic(pred_tab, shade = TRUE, gp = shading_Friendly)
```




```r
# Collapsing occupation to the following based on similarity in the mosaic plots:
# 1, 2 -> "Very Low",    |          4 -> "Middle"
#    3 -> "Low"          |    5, 6, 7 -> "High"

coll.pred_tab <- collapse.table(
  pred_tab,
  Occupation = c("Very Low", "Very Low", "Low", "Middle", "High", "High", "High")
)

pairs(coll.pred_tab, gp = shading_Friendly)
```




```r
mosaic(pred_tab, shade = TRUE, gp = shading_Friendly)
```



Part (c)

Start by fitting a simple model of joint independence of* `Status` *from the predictors, then add additional associations as you consider necessary until you obtain a reasonably well-fitting model.

```
coll_Hoyt <- collapse.table(
  Hoyt,
  Occupation = c("Very Low", "Very Low", "Low", "Middle", "High", "High", "High")
)

(joint <- loglm(~Rank*Occupation*Sex + Status, data = coll_Hoyt))
```

```
## Call:
```

```
## loglm(formula = ~Rank * Occupation * Sex + Status, data = coll_Hoyt)
##
## Statistics:
##                       X^2 df P(> X^2)
## Likelihood Ratio 2520.890 69        0
## Pearson          2512.189 69        0
```

```
(rankstat <- loglm(~Rank*Occupation*Sex + Rank*Status, data = coll_Hoyt))
```

```
## Call:
## loglm(formula = ~Rank * Occupation * Sex + Rank * Status, data = coll_Hoyt)
##
## Statistics:
##                       X^2 df P(> X^2)
## Likelihood Ratio 1458.319 63        0
## Pearson          1469.898 63        0
```

```
(occupstat <- loglm(~Rank*Occupation*Sex + Status*(Rank + Occupation), data = coll_Hoyt))
```

```
## Call:
## loglm(formula = ~Rank * Occupation * Sex + Status * (Rank + Occupation),
##     data = coll_Hoyt)
##
## Statistics:
##                       X^2 df P(> X^2)
## Likelihood Ratio 430.9455 54        0
## Pearson          408.8948 54        0
```

```
(sexstat <- loglm(~Rank*Occupation*Sex + Status*(Rank + Occupation + Sex), data = coll_Hoyt))
```

```
## Call:
## loglm(formula = ~Rank * Occupation * Sex + Status * (Rank + Occupation +
##     Sex), data = coll_Hoyt)
##
## Statistics:
##                       X^2 df     P(> X^2)
## Likelihood Ratio 102.2915 51 2.717215e-05
## Pearson          103.1185 51 2.174678e-05
```

```
(statsexocc <- loglm(~Rank*Occupation*Sex + Status*(Rank + Occupation + Sex) + Status*Sex*Occupation, da
```

```
## Call:
## loglm(formula = ~Rank * Occupation * Sex + Status * (Rank + Occupation +
##     Sex) + Status * Sex * Occupation, data = coll_Hoyt)
##
## Statistics:
##                       X^2 df   P(> X^2)
## Likelihood Ratio 64.11131 42 0.01557704
## Pearson          64.58448 42 0.01411452
```

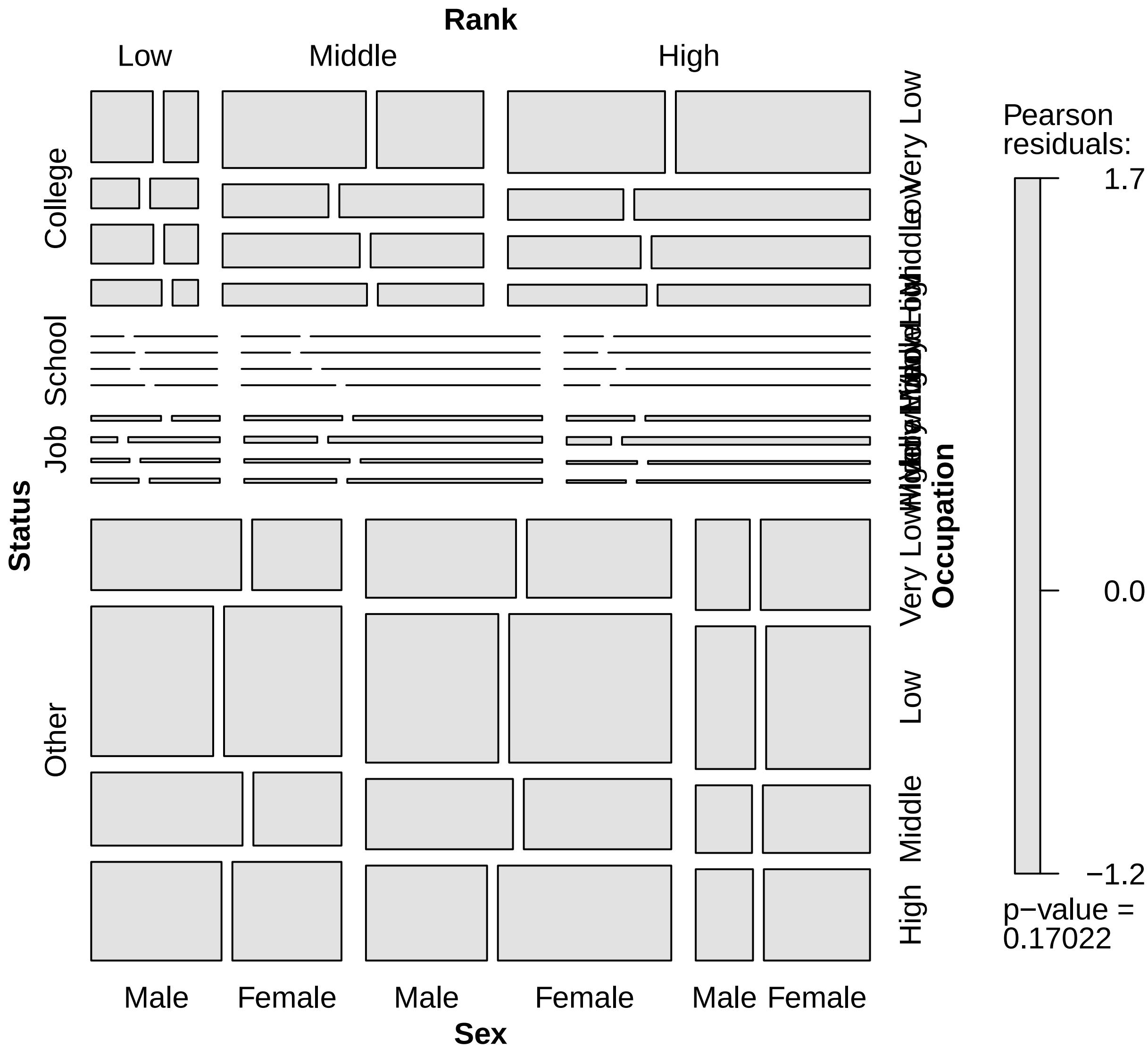
```r
(statsexrank <- loglm(~Rank*Occupation*Sex + Status*(Rank + Occupation + Sex) + Status*(Sex*Occupation +
```

```
## Call:
## loglm(formula = ~Rank * Occupation * Sex + Status * (Rank + Occupation +
##     Sex) + Status * (Sex * Occupation + Occupation * Rank), data = coll_Hoyt)
##
## Statistics:
##                       X^2 df  P(> X^2)
## Likelihood Ratio 30.41883 24 0.1712554
## Pearson          30.45202 24 0.1702186
```

```r
anova(joint, rankstat, occupstat, sexstat, statsexocc, statsexrank)
```

```
## LR tests for hierarchical log-linear models
##
## Model 1:
##  ~Rank * Occupation * Sex + Status
## Model 2:
##  ~Rank * Occupation * Sex + Rank * Status
## Model 3:
##  ~Rank * Occupation * Sex + Status * (Rank + Occupation)
## Model 4:
##  ~Rank * Occupation * Sex + Status * (Rank + Occupation + Sex)
## Model 5:
##  ~Rank * Occupation * Sex + Status * (Rank + Occupation + Sex) + Status * Sex * Occupation
## Model 6:
##  ~Rank * Occupation * Sex + Status * (Rank + Occupation + Sex) + Status * (Sex * Occupation + Occupa
##
##             Deviance df Delta(Dev) Delta(df) P(> Delta(Dev)
## Model 1   2520.89039 69
## Model 2   1458.31891 63 1062.57148         6        0.00000
## Model 3    430.94550 54 1027.37341         9        0.00000
## Model 4    102.29148 51  328.65402         3        0.00000
## Model 5     64.11131 42   38.18017         9        0.00002
## Model 6     30.41883 24   33.69248        18        0.01374
## Saturated    0.00000  0   30.41883        24        0.17126
```

```r
mosaic(statsexrank)
```



Part (g)

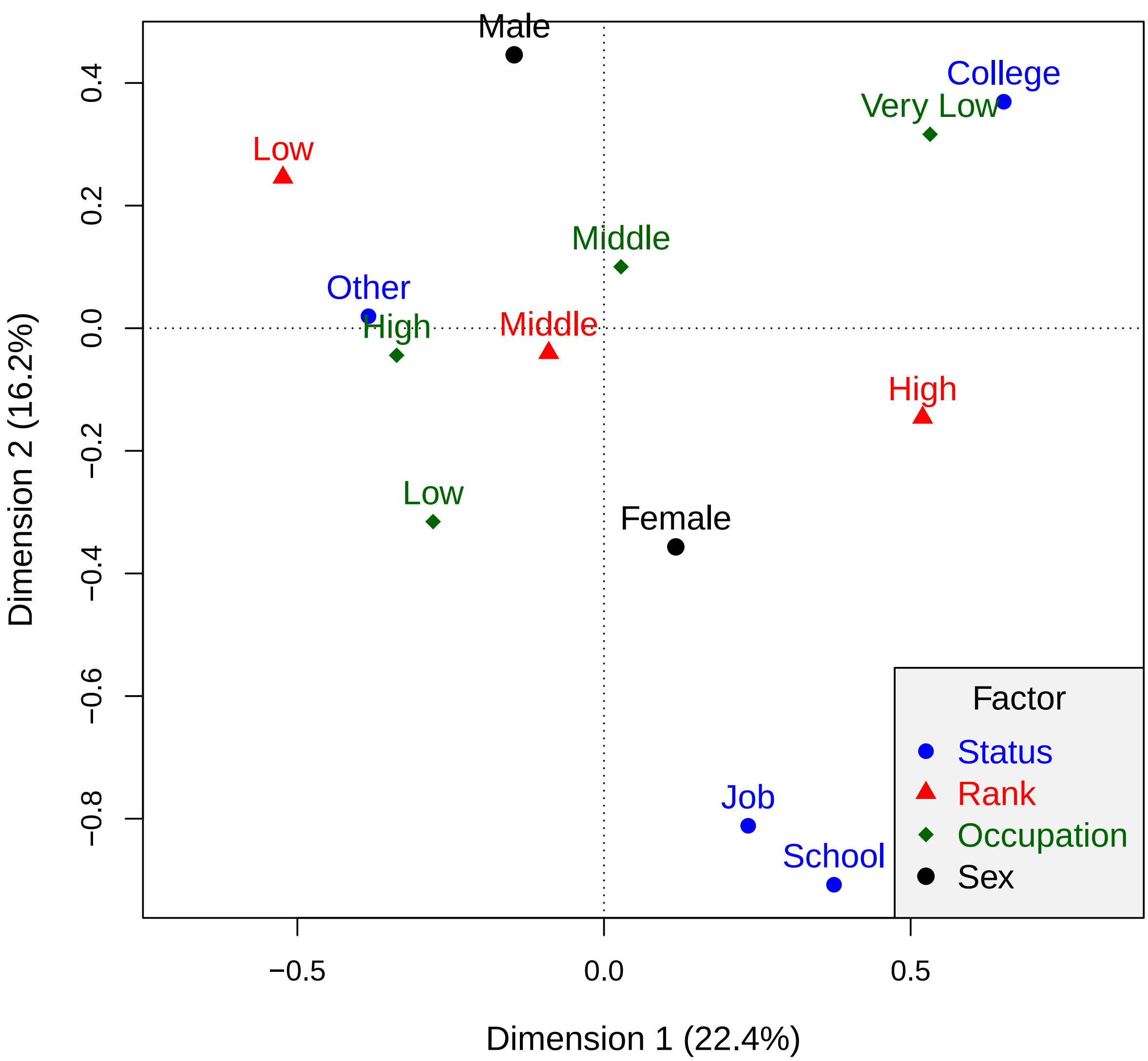
Conduct a multiple correspondence analysis for the `Hoyt` ***data. Provide a reasonable interpretation of the two dimensions that explain the largest proportion of variance.***

```
mjca.coll_Hoyt <- mjca(coll_Hoyt, lambda = "Burt")
summary(mjca.coll_Hoyt)
```

```
##
## Principal inertias (eigenvalues):
##
##  dim    value      %    cum%   scree plot
##  1      0.132576  22.4  22.4   ******
##  2      0.095616  16.2  38.6   ****
```

```
##  3      0.065541  11.1  49.7  ***
##  4      0.062538  10.6  60.3  ***
##  5      0.061990  10.5  70.8  ***
##  6      0.060105  10.2  81.0  ***
##  7      0.048587   8.2  89.2  **
##  8      0.037438   6.3  95.5  **
##  9      0.026417   4.5 100.0  *
##          -------- -----
##  Total: 0.590807 100.0
##
##
## Columns:
##                     name   mass  qlt  inr    k=1 cor ctr    k=2 cor ctr
## 1  |      Status:College |   71  779   86 |  652 590 226 |  369 189 101 |
## 2  |          Status:Job |   22  272   98 |  235  21   9 | -812 251 152 |
## 3  |        Status:Other |  145  721   50 | -384 719 162 |   19   2   1 |
## 4  |       Status:School |   12  194  102 |  375  28  13 | -908 166 105 |
## 5  |           Rank:High |   84  539   77 |  520 500 171 | -146  39  19 |
## 6  |            Rank:Low |   66  450   83 | -523 369 137 |  246  81  42 |
## 7  |         Rank:Middle |  100   26   64 |  -90  22   6 |  -40   4   2 |
## 8  |     Occupation:High |   51  117   86 | -338 115  44 |  -44   2   1 |
## 9  |      Occupation:Low |   80  321   75 | -279 141  47 | -315 180  83 |
## 10 |   Occupation:Middle |   47   10   86 |   28   1   0 |  100   9   5 |
## 11 | Occupation:Very Low |   72  570   82 |  532 421 153 |  316 149  75 |
## 12 |          Sex:Female |  139  667   50 |  117  65  14 | -357 602 185 |
## 13 |            Sex:Male |  111  667   62 | -146  65  18 |  446 602 231 |
```

```r
### BELOW ADAPTED FROM Friendly & Meyer (2016; p. 245), Discrete Data Analysis with R ###
res <- plot(mjca.coll_Hoyt, labels= 0, pch= ".", cex.lab= 1.2)

# extract factor names and levels
coords <- data.frame(res$cols, mjca.coll_Hoyt$factors)
nlev <- mjca.coll_Hoyt$levels.n
fact <- unique(as.character(coords$factor))

cols <- c("blue", "red", "darkgreen", "black")
points(coords[,1:2], pch=rep(16:19, nlev), col=rep(cols, nlev), cex=1.2)
text(coords[,1:2], label=coords$level, col=rep(cols, nlev), pos=3, cex=1.2, xpd=TRUE)

legend("bottomright", legend= c("Status", "Rank", "Occupation", "Sex"), title= "Factor", title.col= "bl
```



Question 2

Student Opinion about the Vietnam War: *The data set* `Vietnam` *in* **vcdExtra** *gives a* $2 \times 5 \times 4$ *contingency table in frequency form reflecting a survey of student opinion on the Vietnam War at the University of North Carolina in May 1967. The table variables are* `sex`, `year` *in school and* `response`, *which has categories: (A) Defeat North Vietnam by widespread bombing and land invasion; (B) Maintain the present policy; (C) De-escalate military activity, stop bombing and begin negotiations; (D) Withdraw military forces Immediately. How does the chosen response vary with* `sex` *and* `year`*?*

Appendix